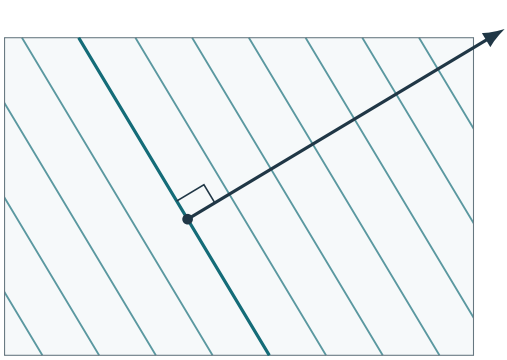

$e^{\mathrm{i}\langle x,\omega\rangle}$

$\langle x, \omega\rangle = c$

constant phase

$$\prod_{\ell=1}^{d} e^{\mathrm{i}x_\ell\omega_\ell} = e^{\mathrm{i}\langle x,\omega\rangle}$$